Mason Miller

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Education

University of Michigan — **Ann Arbor, MI**
Master of Science in Computer Science and Engineering — *August 2026 – May 2027*

University of Michigan — **Ann Arbor, MI**
Bachelor of Science in Computer Science — *August 2022 – May 2026*

- **GPA:** 3.88/4.00
- **Awards:** James B. Angell Scholar, William J. Branstrom Prize, University Honors, Regent's Scholar
- **Relevant Courses:** Data Structures & Algorithms, Operating Systems, Computer Architecture, Web Systems

Technical Skills

Languages: C++, C, Python, ARM, RISC-V, SystemVerilog, Bash
Systems: Linux, XNU, Unix, POSIX, system calls, device drivers
Technologies: GDB, LLDB, QEMU, perf, Valgrind, Git, Make

Experience

EECS 482 Staff — **University of Michigan**
Instructional Assistant (IA) — *August 2025 – Present*

Apple (Core OS) — **Cupertino, CA**
Software Engineer Intern — *May 2025 – August 2025*

- Introduced a new tool that fakes driver architecture and enables user-space testing of kernel storage drivers
- Designed and implemented a fake storage device, automatic driver matching, and a mock infrastructure in C++
- Utilized newly supported tool to increase code coverage of driver data structures and algorithms by over 80%

Qumulo — **Seattle, WA**
Software Engineer Intern — *January 2025 – April 2025*

- Cut minimum cluster cost by 33% by designing and deploying single-node clusters with disk-level fault tolerance
- Built automated performance testing for AWS/Azure to benchmark single-node vs. multi-node cluster performance
- Implemented index caching metrics (hits, misses, gap-fills) and I/O tracking to optimize read path efficiency
- Led read cache DKV system upgrade with safe key remapping, quorum barriers, and RPC-based shard coordination

Ordered Systems Lab — **University of Michigan**
Research Assistant — *May 2024 – December 2024*

- Built a Linux async file reader in C using shared ring buffers and the liburing API for high-throughput I/O
- Benchmarked async I/O libraries (liburing vs. libaio) with Meta RocksDB on a QEMU-based Debian VM

Projects

Out-of-Order RISC-V Processor

- Architected a RISC-V processor using SystemVerilog featuring N-way superscalar out-of-order execution, early tag broadcast, early branch resolution, fast branch recovery, Gshare prediction, branch target buffer, non-blocking data cache, store-load forwarding, and instruction prefetching
- Developed RTL and verification testbenches to validate system performance

Thread Library

- Developed a kernel-level C++ thread library on Unix, managing CPU booting, thread life cycle, and scheduling for multiple CPUs
- Implemented synchronization primitives like spin-locks, mutexes, and condition variables using advanced Unix context management techniques

Virtual Memory Pager

- Designed and implemented a virtual memory pager supporting multiple processes with swap-backed and file-backed memory pages, akin to Unix mmap
- Handled process creation, page faults, memory management unit (MMU) bits, process forking, and destruction with copy-on-write optimization

Multi-threaded Network File Server

- Built a concurrent, crash-consistent network file server, supporting multiple users with nested files and directories
- Ensured crash consistency using committing writes, and optimized concurrency with Boost threads and reader-writer locks. Implemented network communication using POSIX sockets for client-server interactions

Other

Interests: Operating Systems, ML Systems, Rock Climbing, Korean Language, Drum Set, Ethics